Revealing the Milky Way's Center. Image credit: NASA, JPL-Caltech, Susan Stolovy (SSC/Caltech) et al. (2019). https://science.nasa.gov/resource/revealing-the-milky-ways-center/

Hands on astronomical data sonification

Analyzing the universe through sound

Adrián García Riber

agarcia6@esmuc.cat

Final project proposals

1. Sonification of stars, the Tycho-2 Spectral Type Catalog.
2. Sonification of exoplanets discovered in 2026 using the transit detection method.
3. Sonification of potential asteroids detected by the Hubble Space Telescope.

---

1. Sonification of stars. The Tycho-2 Spectral Type Catalog

About stars: https://science.nasa.gov/universe/stars/types/
The Tycho-2 Spectral catalog: https://iopscience.iop.org/article/10.1086/345511

Design and develop a sonification project to represent the Tycho-2 Spectral Type Catalog (Wright+, 2003). You can implement your own synthesizers and/or use any virtual instrument. You can also choose to control them by MIDI or OSC (MIDI recommended). In the following steps you have a mapping strategy proposal but you can design your own approach.

- Use Aladin Sky Atlas to obtain the data from the Tycho-2 Spectral Type Catalog. Export the table to a .txt file and clean the data.
- Map the effective temperature (Teff) to the pitch or fundamental frequency of each note.
- Map the magnitude (VTmag) to the amplitude of each sound or note.
- Map each spectral type (OBAFGKM) to a musical instrument or synthesizer.
- Explore the catalog sequentially. Represent all objects with the same duration, set it to allow the location of the sounds.

2. Sonification of exoplanets discovered in 2026 using the transit detection method.

About exoplanets: https://science.nasa.gov/exoplanets/
NASA Exoplanet Archive:
https://exoplanetarchive.ipac.caltech.edu/cgi-bin/TblView/nph-tblView?app=ExoTbls&config=PS&constraint=default_flag=1&constraint=disc_facility+like+%27%25TESS%25%27

Design and develop a sonification project to represent the exoplanets discovered in the last years using the transit detection method. You can implement your own synthesizers and/or use any virtual instrument. You can also choose to control them by MIDI or OSC (OSC recommended). In the following steps you have a mapping strategy proposal but you can design your own approach.

- Use NASA Exoplanet Catalog to obtain the table with data from the most recently discovered exoplanets. Filter them by the use of the transit detection method in 2026, to obtain a small sample (you can expand it later to represent previous discoveries). Export the table to a .csv file and clean the data.
- Map the Stellar Effective Temperature (st_teff) to the pitch or fundamental frequency of each note.
- Map the Stellar radius (st_rad) to the amplitude of each sound or note.
- Map the orbital period of the planet (pl_orbper) to the oscillation frequency (LFO) of a Tremolo applied to the sonification of the star's brightness.
- Map the planet radius (pl_rade) to the amplitude of the tremolo effect to represent the eclipsed light of the host star.
- Explore the catalog sequentially. Represent all objects with the same duration, set to allow the understanding of the orbital periods.

3. Sonification of potential asteroids detected by the Hubble Space Telescope.

Design and develop a sonification project to represent potential asteroids detected by the Hubble Space Telescope. You can implement your own synthesizers and/or use any virtual instrument. You can also choose to control them by MIDI or OSC. In the following steps you have a mapping strategy proposal but you can design your own approach.

About asteroids: https://science.nasa.gov/solar-system/asteroids/
The catalog: https://cdsarc.cds.unistra.fr/viz-bin/cat/J/A+A/659/A38
Racero 2022: https://www.aanda.org/articles/aa/full_html/2022/03/aa40899-21/aa40899-21.html

- Use Aladin Sky Atlas to obtain the Catalog of potential detections of asteroids (Racero+, 2022) and visualize the Catalog with the list of potential detections from the Hubble Space Telescope. Export the table to a .txt file and clean the data.
- Map the magnitude (Vmag1) to the pitch or fundamental frequency of each note to represent the asteroid.
- Use the actual coordinates of the objects (RA1/DE1, RA2/DE2) to provide a representation of the trajectory of the asteroid within a binaural soundscape.
- Explore the catalog sequentially. Map the duration of the observation (Dur) to the duration of the sound or note.

Report your work

Use this form to deliver your final project. Send a link to a pdf file stored in your own cloud drive. https://forms.gle/KjEYnN2UAxMGiXUS9

Thanks for your participation!

Q&A

agarcia6@esmuc.cat